

Fluid Mechanics: Lecture Notes

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Contents

1	Introduction to Fluid Kinematics	2
2	Units	2
3	Hydrostatics	2

We Finished the chapter about elasticity and now starting the fluid mechanics chapter of the course.

1 Introduction to Fluid Kinematics

Fluid kinematics is the study of how fluids flow and how to describe fluid motion without considering the forces that cause the motion.

- A solid in rest - stable equilibrium (long term order).
- Fluid in rest - Neutral equilibrium (שייט)
 ▶ נוזלים וגזים. Liquids and gasses are in different phase there are differences and similarities.

$$\rho = \rho_0 \quad \text{Incompressible fluid}$$

Consider a water tank in rest. We have internal forces of the kind:

$$dF_i = -pd\vec{S} \quad (1)$$

where $p = p(\vec{x})$ is the pressure. Note: we have only isotropic part and the deviatoric part is vanish.

$$\tau_{ij} = -p\delta_{ij} \quad (2)$$

If theoretically we had deviatoric part it would cause a flow.

$$dF_i = \sigma_{ij}dS_j = -p\delta_{ij}dS_j = -pdS_i \Rightarrow d\vec{F} = -pd\vec{S} \quad (3)$$

2 Units

$$\text{Pressure} = \frac{\text{Force}}{L^2}$$

$$1 \text{ Pascal} = 1 \frac{N}{m^2} \setminus \text{m.k.s}$$

$$1 \text{ Atm} = 1 \frac{\text{Kg} \cdot \text{F}}{\text{cm}^2} = \frac{9.8 \text{ Newton}}{10^{-4} m^2} = 1.01 \times 10^5 \text{ Pascal} \quad (4)$$

$$760 \text{ mm Hg} = 1 \text{ Atm} \quad 1 \text{ Bar} = 10^5 \text{ Pa}$$

$$\text{Psi} = 1 \frac{\text{pound} \times F}{\text{inch}^2} = \frac{0.45 \text{ Kg} \times F}{(2.54)^2 \times \text{cm}^2} = 0.07 \text{ Atm}$$

3 Hydrostatics

Stable equilibrium when $\frac{d^2V}{dx^2}|_{x_0} > 0$ The hydrostatic equation are including the internal force and the external force.

$$\partial_{ij}\sigma_{ij} + \underbrace{\rho g_i}_{f_{\text{ext}}} = 0 \quad (5)$$

Substitute Equation 1

$$\begin{aligned} \partial_{ij}\sigma_{ij} + \underbrace{\rho g_i}_{f_{\text{ext}}} &= -\partial_j(p\delta_{ij}) + \rho g_i \\ &= -\partial_i p + \rho g_i \end{aligned} \quad (6)$$

$$0 = -\vec{\nabla} p + \rho \vec{g} \quad (7)$$